

SHANZAY HUSSAIN

Software Engineering Undergraduate | Full-Stack Developer | AI-Integrated Application Developer

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PROFILE

Software Engineering undergraduate at NED University of Engineering & Technology with hands-on experience building full-stack web applications and integrating AI capabilities into modern software systems. Skilled in developing end-to-end applications using React, TypeScript, Node.js, Express, and Python across cloud platforms. Built applications incorporating LLMs, AI-powered features, retrieval-based systems, and machine learning models. Passionate about AI-integrated full-stack development, intelligent web applications, and shipping practical software from concept to deployment.

EDUCATION

B.E. Software Engineering 2024 – 2028 (Expected)
NED University of Engineering & Technology, Karachi

TECHNICAL SKILLS

Frontend: React.js, Vite, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS
Backend: Node.js, Express.js, FastAPI, Django, REST APIs
Databases: MySQL, PostgreSQL, MongoDB, Firebase Firestore
Programming: Python, JavaScript, TypeScript, C++, C#
AI & Machine Learning: LLM Integration, Prompt Engineering, RAG, Embeddings, NumPy, Pandas, Scikit-learn
Tools & Platforms: Git, GitHub, Supabase, Vercel, Google Colab, Railway, Postman, Figma, Canva, UiPath

PROJECTS

- SmartBudget** — AI-Powered Personal Finance & Spending Risk Platform [GitHub](#) | [Live Demo](#)

React.js · Vite · TypeScript · Express.js · Tailwind CSS · Python · FastAPI · scikit-learn · Firebase · Groq AI (LLaMA 3.3 70B) · Chart.js

 - Built an end-to-end personal finance platform combining traditional budget tracking with machine learning-based overspending risk prediction.
 - Implemented real-time budget analytics, including total spending, percentage of budget used, safe daily spending pace, and category-level tracking.
 - Developed personalized spending insights based on behavioral patterns, including spending trends by mood and day of the week.
 - Integrated a context-aware AI budget assistant using the Groq API and LLaMA 3.3 70B to deliver personalized financial insights from user spending data.
- InterviewIQ** — AI-Powered Mock Interview Platform [GitHub](#) | [Live Demo](#)

React.js · Vite · Express.js · Supabase · Groq AI (LLaMA 3.3 70B) · Chart.js

 - Architected and shipped a full-stack AI interview platform end-to-end — frontend, backend, and cloud deployment.
 - Built an analytics dashboard with Chart.js to track performance trends across sessions, helping users identify skill gaps over time.
 - Implemented speech recognition functionality to simulate realistic interview environments.
- RAG Intelligence Chatbot** — Multimodal AI Knowledge Assistant [GitHub](#) | [Live Demo](#)

React.js · Vite · TypeScript · FastAPI · MongoDB Atlas · ChromaDB · Firebase Auth · OpenAI (GPT-5.4-mini) · fastembed

 - Architected and shipped a full-stack multimodal RAG chatbot supporting document and image uploads, conversational Q&A, and persistent chat history.
 - Built a retrieval pipeline using ChromaDB vector search and local embeddings (fastembed) with similarity-threshold routing to intelligently switch between document-grounded answers and general knowledge queries.
 - Implemented image understanding by integrating a vision model to describe uploads at ingestion time, enabling image content to be retrieved through the same RAG pipeline as text documents.
 - Developed cross-conversation knowledge base search, letting users query across all previously uploaded documents and images regardless of which chat they were shared in.

EXPERIENCE

- Backend Intern (Remote)** — ITVE Pvt Ltd 2 months

FastAPI · MongoDB · JWT · Cloudinary

 - Contributed to the backend of a FastAPI-based REST API with JWT authentication, MongoDB (via Motor) for data persistence, and Cloudinary for image upload handling.

CERTIFICATIONS

Google AI Professional Certificate — Coursera 2026
Covered foundations of AI/ML, prompt engineering, and responsible AI practices.